Applied Volatility Modelling and Dynamic Conditional Correlation

An Empirical Evaluation of GARCH, Asymmetric, and DCC Specifications for Two S&P 500 ETFs on the LSE

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Abstract

This paper evaluates conditional volatility and correlation models for two London Stock Exchange ETFs on the S&P 500: **CSPX.L** (iShares, quoted in US dollars) and **VUAG.L** (Vanguard, quoted in pounds), using $T = 1{,}258$ daily returns from 2021 to 2025. GARCH(1,1) clearly beats EWMA and RiskMetrics, and BIC selects GJR(1,1,1) from 36 GJR specifications for both funds. In this model $\hat{\alpha}_1$ is estimated at zero, so the volatility response is dominated by negative shocks. GJR reacted fastest to the Liberation Day tariff shock (2 April 2025) and, estimated on 2021–2024, gives the best one-day-ahead variance forecasts for 2025 by QLIKE. All Gaussian models leave fat-tailed residuals; Student-t errors ($\nu \approx 5$) remove most of this misfit. A DCC model shows a correlation between 0.44 and 0.93 that falls when the pound/dollar rate is volatile and rose to near its sample maximum just after Liberation Day.

1 Introduction

Volatility clustering is a well-known stylised fact of equity returns, and since volatility cannot be observed it must be measured through a model. Brooks (2019, Chapter 9) discusses several such measures, from exponentially weighted moving averages to GARCH-type models. This paper asks which of them works best in practice.

We compare GARCH(1,1), its restricted versions EWMA and RiskMetrics, and the asymmetric GJR model of Glosten et al. (1993). We judge them by likelihood, information criteria, their reaction to the tariff shock of April 2025, the distribution of their standardised residuals and, because in-sample fit can flatter larger models, by the quality of their out-of-sample forecasts for 2025. As an extension, we study the correlation between the two funds with the dynamic conditional correlation (DCC) model of Engle (2002).

Because both ETFs hold the S&P 500 but are quoted in different currencies, their returns differ by the pound/dollar exchange rate, so their correlation measures how much currency risk matters relative to equity risk.

2 Data

2.1 Asset Selection Methodology

From the LSE ETF universe of the first assignment we selected **CSPX.L** (iShares, US-dollar line) and **VUAG.L** (Vanguard, sterling line): two large, liquid, physically replicating and accumulating S&P 500 funds, so the yearly dividend adjustment of this data set causes no jumps. VUAG.L returns have a correlation of 0.993 with VUSA.L (Vanguard's distributing S&P 500 line) but only 0.57–0.58 with FTSE 100 ETFs. The price file has 1,290 dates in 2021–2025; we drop the 31 dates without prices, which leaves 1,259 trading days.

2.2 Returns and Descriptive Statistics

We compute daily log-returns in percent, $y_t = 100\,(\ln P_t - \ln P_{t-1})$, which gives $T = 1{,}258$ returns from 5 January 2021 to 30 December 2025. As the assignment asks, the returns are stored in a separate file (`returns.csv`), and all further analysis uses only this file.

Since both funds hold the same index, the difference of their returns is approximately the change in the exchange rate, $y_t^{\mathrm{CSPX}} - y_t^{\mathrm{VUAG}} \approx \Delta \ln(\text{USD per GBP})_t$. This difference has a standard deviation of 0.54% per day. Its largest values all occur during the UK 'mini-budget' crisis and the following sterling swings of autumn 2022, from -3.05% on 23 September to $+2.67\%$ on 1 December 2022. The return correlation is 0.85 over the full sample.

Table 1 shows a small positive mean, negative skewness and excess kurtosis of about 3 for both funds. Normality is rejected, and the Ljung–Box test on squared returns shows strong volatility clustering, which is also visible in Figure 1. VUAG.L is less volatile than CSPX.L (14.6% against 16.1% per year). The dollar tends to strengthen when US equities fall: the correlation between the CSPX.L return and the currency difference above is 0.44. For a sterling investor, the currency therefore softens equity losses.

Table 1: Descriptive summary statistics of daily log-returns (%), 2021–2025, $T = 1{,}258$

	Mean	Std. dev.	Ann. vol.	Skewness	Exc. kurt.	Min	Max	JB p	$\mathrm{LB}_{10}(y^2)$ p
CSPX.L	0.054	1.014	16.1	−0.33	2.71	−4.98	5.11	<0.001	<0.001
VUAG.L	0.055	0.917	14.6	−0.34	3.01	−4.89	3.72	<0.001	<0.001

Ann. vol.: standard deviation $\times\sqrt{252}$ in %. Exc. kurt.: excess kurtosis. JB: Jarque–Bera test of normality. $\mathrm{LB}_{10}(y^2)$: Ljung–Box test with 10 lags on squared returns.

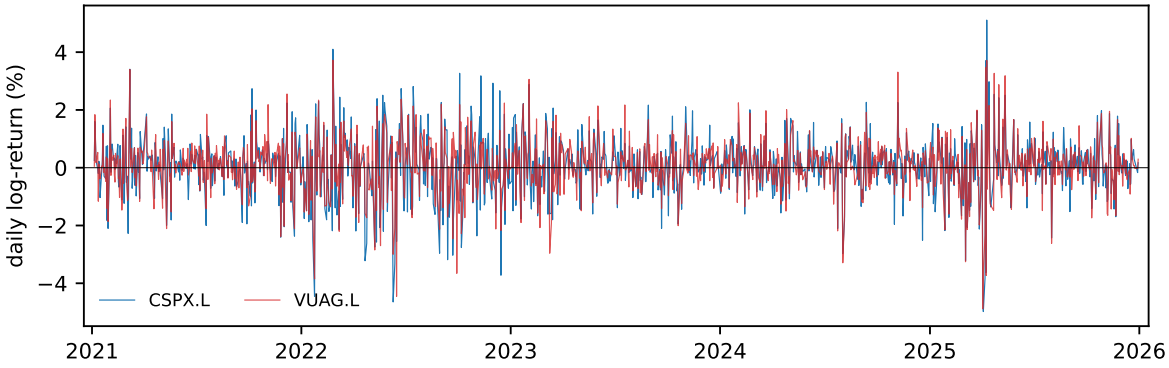


Figure 1: Daily log-returns of CSPX.L (US dollar) and VUAG.L (pound sterling).

3 Methodology

3.1 Baseline Volatility Models

All models share the observation equation $y_t \sim \mathcal{N}(\mu, \sigma_t^2)$ with $u_t = y_t - \mu$. We use the notation of the assignment:

1. **GARCH(1,1)**:
$$\sigma_{t+1}^2 = \omega + \alpha u_t^2 + \delta \sigma_t^2. \tag{1}$$

2. **Exponentially Weighted Moving Average (EWMA)**: the restriction $\omega = 0$, $\alpha = 1 - \delta$ gives
$$\sigma_{t+1}^2 = (1-\delta)\, u_t^2 + \delta \sigma_t^2, \tag{2}$$
with free parameters μ and δ. The variance is integrated and has no long-run level.

3. **RiskMetrics**: in addition $\delta = 0.94$ and $\mu = \bar{y}$, so nothing is optimised numerically:
$$\sigma_{t+1}^2 = 0.06\, u_t^2 + 0.94\, \sigma_t^2. \tag{3}$$
The mean is still estimated from the data (by $\bar{y}$ instead of by maximum likelihood), so we count it as one parameter in the information criteria and in the LR test against EWMA.

3.2 Advanced Model: GJR-GARCH

As the advanced model we use the GJR model (Glosten et al., 1993),

$$\sigma_{t+1}^2 = \omega + \sum_{j=1}^{q} \alpha_j u_{t-j+1}^2 + \sum_{i=1}^{p} \delta_i \sigma_{t-i+1}^2 + \sum_{k=1}^{o} \gamma_k u_{t-k+1}^2 \, \mathbb{I}_{\{u_{t-k+1}<0\}}, \tag{4}$$

for all 36 combinations of $p \in \{0,1,2,3\}$, $q \in \{1,2,3\}$ and $o \in \{0,1,2\}$. With $\gamma_k > 0$ a negative return raises the next variance more than a positive return of the same size (the leverage effect). GARCH(1,1) is the special case GJR(1,1,0).

As a robustness check we re-estimate the selected GJR model with Student-t errors, $u_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \sqrt{\nu/(\nu-2)} \sim t_\nu$, so that ε_t has unit variance and $\nu > 2$ measures the tail thickness. The normal model is the limit $\nu \to \infty$.

3.3 Estimation and Model Comparison

All models are estimated by maximum likelihood with the Gaussian log-likelihood

$$\log L(\theta) = -\tfrac{1}{2}\sum_{t=1}^{T}\Big(\ln 2\pi + \ln\sigma_t^2 + u_t^2/\sigma_t^2\Big). \tag{5}$$

The recursion starts at $\sigma_1^2 = T^{-1}\sum_t u_t^2$, and pre-sample values are set to the same value. We impose $\omega > 0$, $\alpha_j, \delta_i \geq 0$, a non-negative response to negative shocks at each lag ($\alpha_k + \gamma_k \geq 0$) and $\sum\alpha_j + \sum\delta_i + \frac{1}{2}\sum\gamma_k < 1$. The optimiser (SLSQP) starts from several values, including the optimum of every model nested one order lower. This ensures in practice that a larger model never fits worse than a model nested in it. We work with returns in percent, because with raw returns ω is of order 10^{-5}, which makes the numerical optimisation badly scaled.

Standard errors come from a numerical Hessian. Normality is clearly rejected (Section 4.5), so we treat the estimates as quasi-maximum likelihood and report the robust sandwich standard errors of Bollerslev and Wooldridge (1992). When a parameter lies on the boundary of the parameter space (such as $\hat{\alpha}_1 = 0$ below), its standard error and the usual χ^2 distribution of likelihood ratio (LR) tests are not valid, and we interpret them with care. Moreover, because the Gaussian likelihood is misspecified, LR statistics and information criteria are only approximately valid under QML. We therefore check the main LR conclusions with robust Wald t-statistics, which lead to the same conclusions: the asymmetry γ_1 in GJR(1,1,1) has robust $t = 4.7$ (CSPX.L) and 4.2 (VUAG.L), while $(\hat{\delta}_1 - 0.94)$/s.e. in EWMA is only 0.33 and 0.58.

To compare models we use LR tests for nested models (EWMA within GARCH, RiskMetrics within EWMA, GARCH within GJR) and the information criteria AIC $= -2\log L + 2k$ and BIC $= -2\log L + k\ln T$ (lower is better; BIC is the SBIC of Brooks, 2019), which penalise extra parameters. With $T =$ 1,258, BIC charges about 7 points per parameter against 2 for AIC, so it favours smaller models; we use BIC for the final choice and AIC as a check. We also check whether the standardised residuals $v_t = u_t/\sigma_t$ behave as the model implies.

3.4 Out-of-Sample Forecast Evaluation

We estimate every model once on 2021–2024 (1,007 returns), re-selecting the GJR order by BIC on this period only, and keep the parameters fixed. Running the recursion through 2025 then gives one-day-ahead forecasts σ_t^2 for the 251 trading days of 2025 that use only returns up to $t-1$. Since the true variance is not observed, we use y_t^2 as proxy and evaluate the forecasts with the QLIKE loss $\ln\sigma_t^2 + y_t^2/\sigma_t^2$ and the squared error $(y_t^2 - \sigma_t^2)^2$. QLIKE is robust to the noise in the proxy and penalises under-prediction more heavily (Patton, 2011). Equal predictive accuracy against GJR is tested with the Diebold and Mariano (1995) test, using a Newey–West variance with five lags. We also report the predictive log-likelihood and the share of days with $|u_t| > 2\sigma_t$, which should be about 4.6% under normality.

3.5 Dynamic Conditional Correlation (DCC)

Using the standardised residuals $v_t = (v_{1t}, v_{2t})'$ of the selected GJR models and $S = \mathrm{var}(v)$, we estimate the correlation part of the DCC(1,1) model of Engle (2002):

$$Q_{t+1} = S\,(1 - a - b) + a\,v_t v_t' + b\,Q_t, \qquad Q_1 = S, \tag{6}$$

$$R_{t+1} = (\mathrm{diag}\,Q_{t+1})^{-1/2}\,Q_{t+1}\,(\mathrm{diag}\,Q_{t+1})^{-1/2}, \tag{7}$$

and maximise $\sum_t \log\phi_2(v_t; 0, R_t)$ over (a, b). The assignment writes Q_t in the middle of the expression for R_{t+1}; we use Q_{t+1}, so that R_{t+1} is a proper correlation matrix. Here a measures the reaction of the correlation to new shocks and b its persistence. The correlation R_t used to evaluate v_t depends only on $v_1, \ldots, v_{t-1}$ (given S and the first-step GJR estimates, which use the full sample).

4 Univariate Results

4.1 Choice of the GJR Model

Table 2 lists the five best GJR models by BIC; Table 9 in the appendix gives all 36. The lagged variance is the most important ingredient. Without it ($p = 0$, ARCH models) the fit is much worse, and even

three ARCH lags fall far short of a single δ_1. The asymmetry term comes next: moving from $o = 0$ to $o = 1$ raises the log-likelihood by about 34 (CSPX.L) and 32 (VUAG.L) points. Additional ARCH lags ($q > 1$) add almost nothing. Once the asymmetry term is included their estimates go to zero, so many of the larger models reproduce the likelihood of a smaller one and are penalised for the extra parameters.

BIC selects GJR(1,1,1) for both series. AIC prefers GJR(2,1,1) for CSPX.L and GJR(3,1,2) for VUAG.L, but only by 0.2 points. LR tests do not reject GJR(1,1,1) against these models ($p = 0.14$ and $p = 0.10$). We therefore use GJR(1,1,1) for both series, which also gives one specification for the DCC model.

Table 2: The five best GJR(p, q, o) models by BIC, and GARCH(1,1)

	CSPX.L					VUAG.L				
Rank	(p, q, o)	$\log L$	AIC	BIC	ΔBIC	(p, q, o)	$\log L$	AIC	BIC	ΔBIC
1	(1,1,1)	−1649.5	3309.0	3334.7	0.0	(1,1,1)	−1560.6	3131.2	3156.9	0.0
2	(2,1,1)	−1648.4	3308.8	3339.7	5.0	(2,1,1)	−1560.6	3133.1	3163.9	7.1
3	(1,1,2)	−1649.5	3311.0	3341.8	7.1	(1,1,2)	−1560.6	3133.2	3164.0	7.1
4	(1,2,1)	−1649.5	3311.0	3341.8	7.1	(1,2,1)	−1560.6	3133.2	3164.0	7.1
5	(2,1,2)	−1648.3	3310.7	3346.6	11.9	(3,1,1)	−1559.2	3132.5	3168.4	11.5
GARCH	(1,1,0)	−1683.7	3375.3	3395.9	61.2	(1,1,0)	−1592.6	3193.3	3213.8	57.0

p: lags of σ_t^2, q: lags of u_t^2, o: asymmetric lags. ΔBIC: difference with the best model. All 36 models are in Table 9 in the appendix.

4.2 Estimation Results and Model Selection

Table 3 reports the estimates, robust standard errors, log-likelihoods and information criteria of the five models, and Table 4 the LR tests.

The mean return is about 0.05–0.09% per day. For GARCH(1,1), $\hat{\alpha} = 0.13$ (CSPX.L) and 0.10 (VUAG.L) and $\hat{\delta} = 0.83$ and 0.86. Persistence is about 0.96, so a shock to the variance has a half-life of about 15 trading days. The implied unconditional variance $\hat{\omega}/(1 - \hat{\alpha} - \hat{\delta})$ equals 1.04 and 0.86, close to the sample variances.

The EWMA estimates of δ are 0.946 and 0.952, close to the RiskMetrics value 0.94, and the LR test does not reject RiskMetrics against EWMA. The restriction that turns GARCH into EWMA, however, is strongly rejected (LR of 47 and 46 with two degrees of freedom). The data prefer a variance that reverts to a long-run level and reacts more strongly to new shocks.

For both assets **GJR(1,1,1)** gives the highest log-likelihood and the lowest AIC and BIC of the Gaussian models in Table 3. Its $\hat{\alpha}_1$ is zero in both series, while $\hat{\gamma}_1 = 0.217$ and 0.195 with robust standard errors of 0.05. The volatility response is thus dominated by falling prices, a strong form of the leverage effect. Because $\hat{\alpha}_1$ lies on the boundary its standard error is not reliable, and a small positive α_1 cannot be ruled out. Still, the asymmetry is convincing: GJR improves on GARCH(1,1) by about 34 (CSPX.L) and 32 (VUAG.L) log-likelihood points for one extra parameter.

Most parameters are precisely estimated. The standard errors of the GJR columns (marked † in Table 3) are affected by the boundary estimate of α_1, which matters mainly for CSPX.L: ω has a robust $t \approx 1.6$ here, but $t \approx 3$ with α_1 fixed at zero (Section 8).

The last column re-estimates GJR(1,1,1) with Student-t errors. The estimated degrees of freedom are $\hat{\nu} = 5.3$ and 5.2, far from the normal limit, and the log-likelihood rises by 31 and 38 points for one extra parameter. The variance dynamics hardly change ($\hat{\alpha}_1 = 0$, $\hat{\delta}_1 = 0.84$, $\hat{\gamma}_1 = 0.27$ and 0.21), so the asymmetry is not an artefact of the normal assumption. For CSPX.L the robust standard errors of the t model are large (for example 0.29 for $\hat{\delta}_1$, against 0.09 non-robust). This is a consequence of $\hat{\alpha}_1 = 0$ lying on the boundary of the parameter space, which makes the numerical standard errors unreliable (Section 8).

4.3 Full Sample Volatility Dynamics

Figure 2 plots the full-sample $2\sigma_t$ bands of the four models against the absolute returns $|y_t|$. All models trace the main regimes: the 2022 bear market, the calm of 2023–2024, the short spike of August 2024 and the tariff shock of 2025. They differ in timing and amplitude. GJR has the sharpest peaks after market falls, and in rising markets it often gives the lowest volatility. EWMA and RiskMetrics are the smoothest. They react weakly to a single large return ($\alpha \approx 0.05$), but stay elevated for a long time because they lack

Table 3: Parameter estimates and model comparison

CSPX.L					
	GARCH(1,1)	EWMA	RiskMetrics	GJR(1,1,1)†	GJR(1,1,1)-t†
μ	0.088	0.075	0.054	0.049	0.081
	(0.023)	(0.030)	(0.029)	(0.024)	(0.026)
ω	0.0470	*0*	*0*	0.0375	0.0330
	(0.0167)			(0.0235)	(0.0598)
α_1	0.127	*0.054*	*0.060*	0.000	0.000
	(0.031)			(0.083)	(0.246)
δ_1	0.828	0.946	*0.940*	0.853	0.841
	(0.039)	(0.017)		(0.099)	(0.293)
γ_1				0.217	0.266
				(0.046)	(0.147)
ν					5.26
					(2.53)
Persistence	0.955	1.000	1.000	0.962	0.974
k	4	2	1	5	6
$\log L$	−1683.7	−1707.2	−1707.8	−1649.5	−1618.9
AIC	3375.3	3418.3	3417.5	3309.0	3249.9
BIC	3395.9	3428.6	3422.7	3334.7	3280.7
VUAG.L					
	GARCH(1,1)	EWMA	RiskMetrics	GJR(1,1,1)†	GJR(1,1,1)-t†
μ	0.087	0.078	0.055	0.055	0.078
	(0.024)	(0.029)	(0.026)	(0.022)	(0.020)
ω	0.0371	*0*	*0*	0.0426	0.0445
	(0.0133)			(0.0129)	(0.0180)
α_1	0.100	*0.048*	*0.060*	0.000	0.000
	(0.023)			(0.024)	(0.019)
δ_1	0.857	0.952	*0.940*	0.850	0.841
	(0.028)	(0.020)		(0.033)	(0.046)
γ_1				0.195	0.211
				(0.046)	(0.058)
ν					5.21
					(0.73)
Persistence	0.957	1.000	1.000	0.947	0.947
k	4	2	1	5	6
$\log L$	−1592.6	−1615.4	−1616.9	−1560.6	−1523.0
AIC	3193.3	3234.9	3235.8	3131.2	3058.0
BIC	3213.8	3245.2	3241.0	3156.9	3088.8

Robust (sandwich) standard errors in parentheses. Values in italics are fixed by the model restrictions; for RiskMetrics $\mu = \bar{y}$, and the reported standard error is that of the sample mean. Persistence: $\sum \alpha_j + \sum \delta_i + \frac{1}{2}\sum \gamma_k$. GJR(1,1,1)-$t$: Student-$t$ errors with ν degrees of freedom. †: $\hat{\alpha}_1 = 0$ lies on the boundary of the parameter space, so the standard errors in this column are unreliable (see Section 8).

Table 4: Likelihood ratio tests of nested models

	Restricted vs unrestricted	LR	df	p
CSPX.L	EWMA vs GARCH(1,1)	47.01	2	<0.001
	RiskMetrics vs EWMA	1.19	1	0.274
	GARCH(1,1) vs GJR(1,1,1)	68.31	1	<0.001
VUAG.L	EWMA vs GARCH(1,1)	45.61	2	<0.001
	RiskMetrics vs EWMA	2.92	1	0.087
	GARCH(1,1) vs GJR(1,1,1)	64.10	1	<0.001

p-values from the χ^2_{df} distribution. EWMA vs GARCH: the restriction $\omega = 0$ lies on the boundary of the parameter space; GARCH vs GJR: $\hat{\alpha}_1 = 0$ lies on the boundary. Both p-values are therefore approximate, but the statistics are far beyond any critical value. RiskMetrics vs EWMA: one restriction ($\delta = 0.94$). Because $\bar{y}$ is not the ML estimate of μ, the statistic is slightly too large; with μ at its ML value it is 0.51 ($p = 0.47$) and 1.75 ($p = 0.19$), with the same conclusion.

mean reversion. Their peaks are generally lower than those of GJR, and around Liberation Day they also came later (11 April, against 8–10 April for GJR): a single shock adds only about 5–6% of u_{t-1}^2 to the variance, so EWMA and RiskMetrics only rise strongly after several large returns in a row, whereas GJR adds about 20% of u_{t-1}^2 after a negative return ($\hat{\gamma}_1 \approx 0.2$). GARCH lies in between. The share of days with $|u_t| > 2\sigma_t$ (nominal 4.6% under normality) broadly supports this ranking: 5.3% and 5.5% for GJR, 5.8% and 6.0% for GARCH, 6.4% and 6.0% for EWMA, and 6.7% and 6.2% for RiskMetrics (CSPX.L and VUAG.L). All models exceed the nominal rate somewhat, as expected with fat tails, but GJR the least.

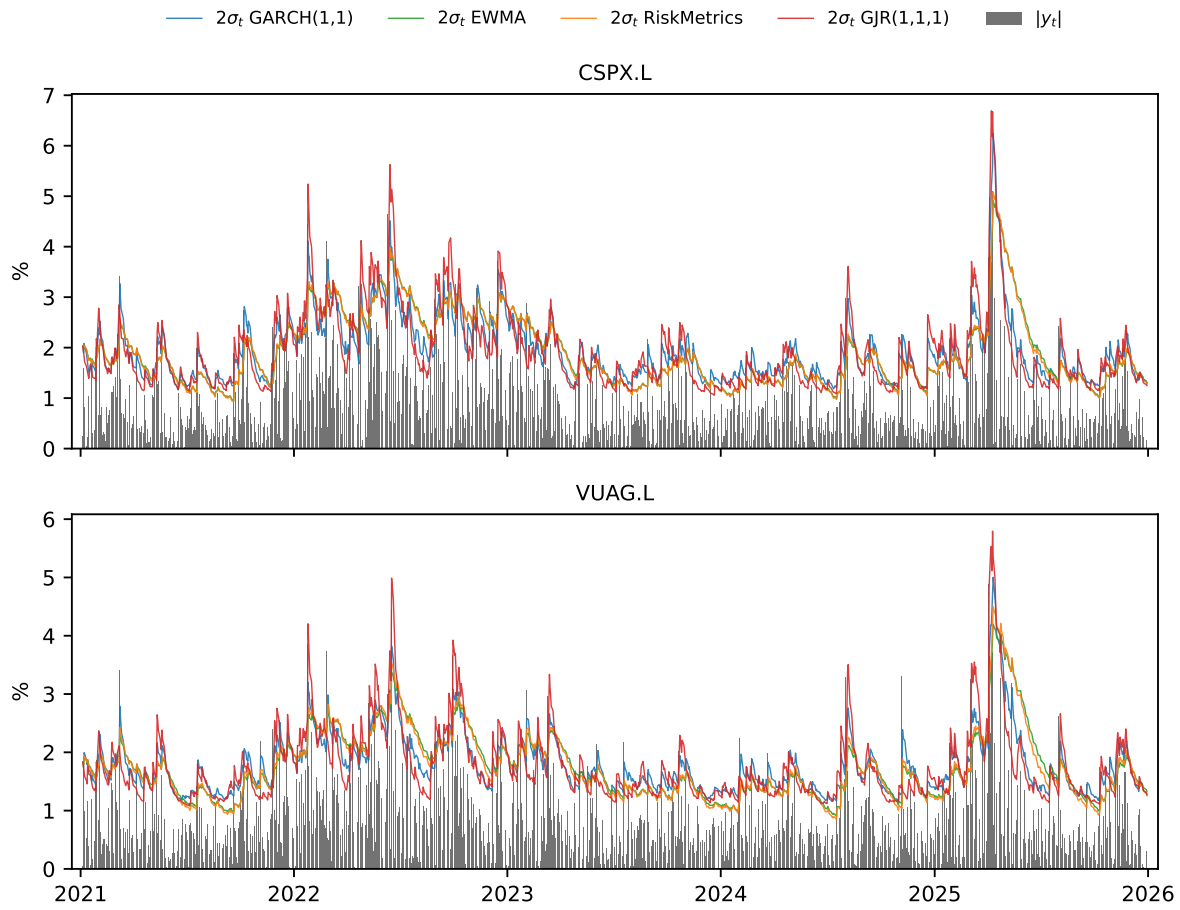


Figure 2: Full sample volatility dynamics: $2\sigma_t$ of the four models against absolute returns $|y_t|$ (grey).

4.4 Reaction to the Tariff Shock around Liberation Day (April 2, 2025)

The tariffs were announced on 2 April 2025, after the London close. Figure 3 zooms in on this period. For the first London return after the announcement (3 April) all models predicted $2\sigma_t \approx 2.1$–2.3%, but the returns were -3.8% (CSPX.L) and -4.9% (VUAG.L). VUAG.L fell more because the dollar weakened as well, so here the dollar did not act as a safe haven.

The next day GJR raised $2\sigma_t$ to 4.2% and 4.8%, GARCH to 3.5% and 3.7%, and EWMA only to 2.8% and 3.0%. GJR peaked at 6.7% and 5.8% within a week, close to the realised absolute returns of 4–5%; EWMA and RiskMetrics peaked at only 4.2–5.1%, and only on 11 April. The rebound of up to 5% after the tariff pause (9 April) barely moves GJR, whose volatility falls from 11 April while GARCH still rises. By early May GJR is back near its pre-shock level, whereas EWMA and RiskMetrics still give $2\sigma_t \approx 3.6\%$. GJR thus followed both the build-up and the decay of the shock most closely.

4.5 Residual Diagnostics via Normal Quantile Transform

If a model is correctly specified, $v_t = u_t/\sigma_t \sim \mathcal{N}(0,1)$ and the normal quantile transform $q_t = \Phi(v_t)$ is uniform on $(0,1)$. (We use Φ, the probability integral transform; Φ^{-1} as written in the assignment would

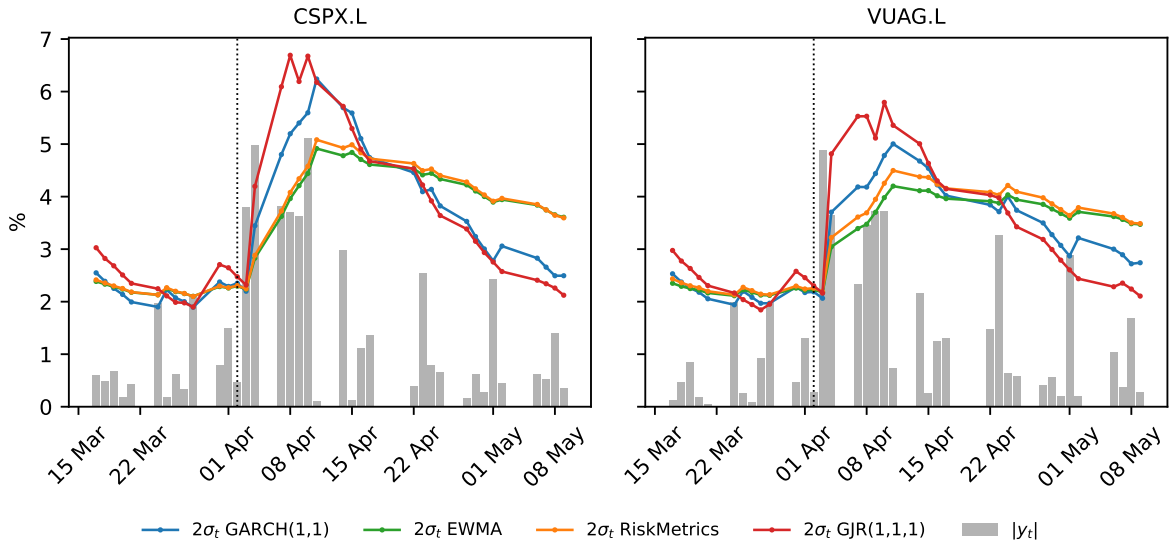


Figure 3: Volatility reaction ($2\sigma_t$) around Liberation Day (dotted line, 2 April 2025), 17 March – 9 May 2025.

map to the real line instead.) For the Student-t model Φ is replaced by the cdf of the standardised $t_{\hat{\nu}}$ distribution. Figure 4 shows the histograms of q_t, and Table 5 reports formal tests.

The volatility dynamics are captured well. No model leaves significant autocorrelation in v_t^2, and GARCH and GJR give v_t a standard deviation of 1.00. EWMA and RiskMetrics give 1.04–1.05, because their variance is not tied to the level of the data. The shape of the distribution, however, is not normal. The histograms have too many observations in the middle and in the lowest bin, and too few at moderate quantiles and in the upper tail. This is the signature of fatter tails than the normal. The excess kurtosis of v_t is 1.3–2.6, and both the Kolmogorov–Smirnov and the χ^2 test reject uniformity for every Gaussian model. The lowest bin is higher than the highest one, reflecting the remaining negative skewness. GJR reduces the kurtosis most (1.34 and 1.77) but does not remove it.

With Student-t errors the histograms become nearly flat (last column of Figure 4). The KS test no longer rejects ($p = 0.40$ and 0.94), and for VUAG.L neither does the χ^2 test ($p = 0.43$). For CSPX.L the χ^2 test still rejects ($p = 0.002$): the lowest bin is too full, because the symmetric t cannot capture the skewness of -0.5. The fat tails are therefore the main failure of the Gaussian models; a skewed-t distribution would be the natural next step.

Table 5: Diagnostics of the standardised residuals v_t and of $q_t = \Phi(v_t)$

	Model	Std. dev. v_t	Skew.	Exc. kurt.	JB p	$\text{LB}_{10}(v^2)$ p	KS p	χ^2_9 p
CSPX.L	GARCH(1,1)	0.999	−0.45	1.40	<0.001	0.748	<0.001	<0.001
	EWMA	1.039	−0.49	1.70	<0.001	0.234	0.005	<0.001
	RiskMetrics	1.044	−0.51	1.76	<0.001	0.481	<0.001	<0.001
	GJR(1,1,1)	1.000	−0.51	1.34	<0.001	0.421	<0.001	<0.001
	GJR(1,1,1)-t	0.984	−0.54	1.48	<0.001	0.266	0.404	0.002
VUAG.L	GARCH(1,1)	1.000	−0.29	2.10	<0.001	0.726	0.004	<0.001
	EWMA	1.041	−0.23	2.48	<0.001	0.381	0.017	<0.001
	RiskMetrics	1.052	−0.21	2.64	<0.001	0.718	0.001	<0.001
	GJR(1,1,1)	1.001	−0.32	1.77	<0.001	0.518	0.002	<0.001
	GJR(1,1,1)-t	0.989	−0.32	1.78	<0.001	0.456	0.935	0.428

JB: Jarque–Bera test of normality of v_t. $\text{LB}_{10}(v^2)$: Ljung–Box test (10 lags) on v_t^2. KS: Kolmogorov–Smirnov test of $q_t \sim U(0,1)$. χ^2_9: Pearson test of equal counts in 10 bins of q_t. Both tests ignore that the parameters are estimated, so their p-values are only approximate. For GJR(1,1,1)-t, q_t uses the Student-t cdf; JB still tests normality of v_t.

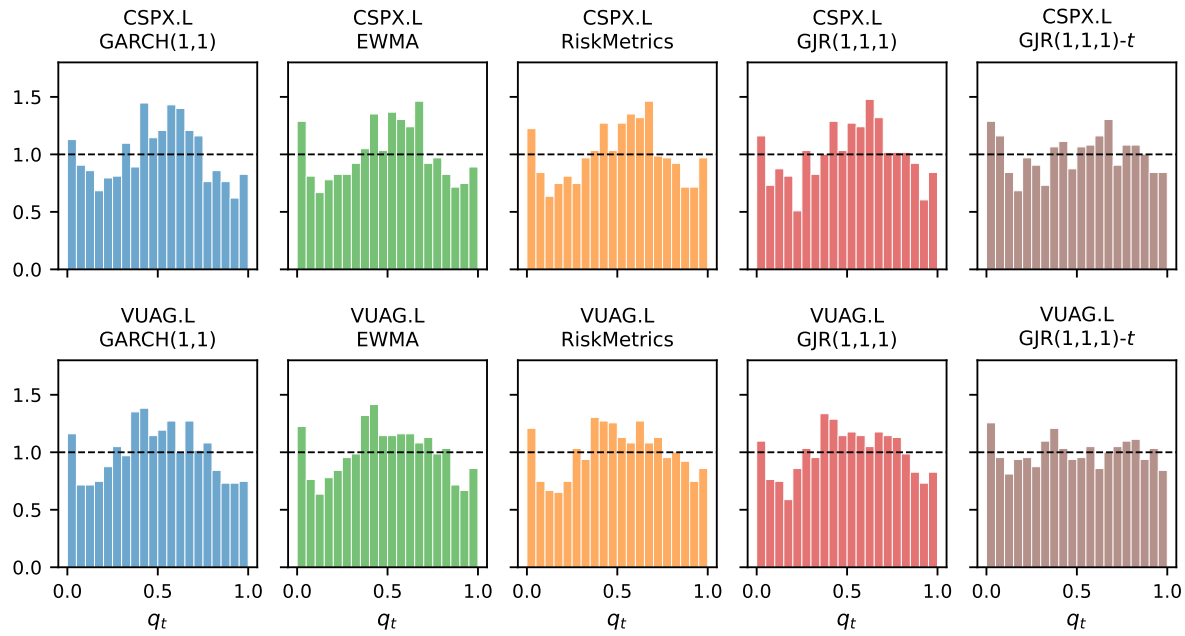


Figure 4: Histograms of q_t (20 bins) for the four Gaussian models and GJR(1,1,1) with Student-t errors, against the uniform $U(0,1)$ density (dashed).

5 Out-of-Sample Forecasts for 2025

Estimated on 2021–2024 alone, BIC again selects GJR(1,1,1) for both series ($\hat{\alpha}_1 = 0$, $\hat{\gamma}_1 \approx 0.2$), so the model choice is stable over time. Table 6 shows that the in-sample ranking largely holds out of sample: the two GJR models have the lowest QLIKE, followed by GARCH, RiskMetrics and EWMA. GJR is significantly better than EWMA and RiskMetrics for both funds ($p \leq 0.02$) and than GARCH for CSPX.L ($p = 0.02$), but not for VUAG.L ($p = 0.40$). April 2025 contributes much of the gain; without it EWMA and RiskMetrics are still worse, with $p \approx 0.01$–0.02 (CSPX.L) and $p = 0.05$–0.06 (VUAG.L). The MSE gives a broadly similar ranking, but is dominated by a few very large y_t^2 and separates none of them.

EWMA puts $|u_t|$ outside $2\sigma_t$ on 8.0% of the days for both funds, significantly more than the nominal 4.6% (binomial $p = 0.015$), because it reacts too slowly when a shock starts; GJR and GARCH stay close to the nominal rate. The Student-t model gives practically the same variance forecasts as the normal GJR, but a much higher predictive log-likelihood (+9 and +14), because it puts more mass in the tails. For volatility forecasts the normal QMLE thus suffices; for risk measures such as value-at-risk the t distribution is preferable.

Table 6: One-day-ahead forecasts for 2025 (251 days) from models estimated on 2021–2024

	Model	QLIKE	DM p	MSE	DM p	Pred. log L	$\lvert u_t\rvert > 2\sigma_t$
CSPX.L	GARCH(1,1)	0.818	0.021	7.44	0.208	−333.6	4.8%
	EWMA	0.968	0.009	8.75	0.197	−352.3	8.0%
	RiskMetrics	0.908	0.004	8.34	0.154	−344.5	7.2%
	GJR(1,1,1)	0.748	–	6.07	–	−324.7	4.4%
	GJR(1,1,1)-t	0.755	0.512	6.06	0.963	−315.9	4.4%
VUAG.L	GARCH(1,1)	0.820	0.398	5.92	0.273	−333.6	6.0%
	EWMA	0.978	0.018	6.65	0.189	−353.9	8.0%
	RiskMetrics	0.898	0.020	6.27	0.143	−342.9	6.4%
	GJR(1,1,1)	0.788	–	5.25	–	−329.3	5.2%
	GJR(1,1,1)-t	0.788	0.970	5.29	0.306	−315.2	5.2%

QLIKE and MSE: mean loss with y_t^2 as variance proxy (lower is better; the mean return of about 0.05% is negligible relative to $\sigma_t \approx 1\%$). DM p: Diebold–Mariano test of equal loss with GJR(1,1,1). Pred. log L: sum of the one-step-ahead log-densities. Last column: share of days with $|u_t| = |y_t - \hat{\mu}| > 2\sigma_t$ (nominal 4.6% under normality, 5.0% under the fitted t).

6 Extra: Correlation and DCC Analysis

6.1 Sub-Sample Correlations

Table 7 reports the correlation between the standardised GJR(1,1,1) residuals over the full sample and its three thirds. The full-sample correlation is 0.81. It falls from 0.85 in the first third to 0.74 in the second third (September 2022 – May 2024) and recovers to 0.84 in the last third. With standard errors of 0.01–0.02 these differences are large, and a Fisher-z test rejects equal correlations ($\chi^2_2 = 22.7$, $p < 0.001$), so the correlation is not stable.

The currency is the most likely explanation. The second third contains the sterling crisis of autumn 2022 and the large dollar moves of 2023. The standard deviation of the return difference is 0.65% there, against 0.51% and 0.43% in the other thirds, and equity volatility was lower than in the first third. With relatively more currency noise, the two returns move less closely together. The correlation of the raw returns is higher (0.85 over the full sample) because both volatilities move together: raw returns give most weight to turbulent days, when the common equity shock dominates, while standardisation gives every day equal weight.

Table 7: Standardised residual correlation breakdown

Period	Dates	T	$\hat{\rho}(v_{1t}, v_{2t})$	s.e.	$\hat{\rho}(y_{1t}, y_{2t})$
Full sample	2021-01-05 – 2025-12-30	1258	0.810	0.010	0.849
First third	2021-01-05 – 2022-09-02	420	0.853	0.013	0.893
Second third	2022-09-05 – 2024-05-02	419	0.742	0.022	0.738
Last third	2024-05-03 – 2025-12-30	419	0.835	0.015	0.895
Test of equal correlation in the three thirds: $\chi^2_2 = 22.7$, $p < 0.001$					

s.e.: asymptotic standard error $(1-\hat{\rho}^2)/\sqrt{T}$, which assumes normality and is somewhat too small given the fat tails of v_t. The last column gives the correlation of the raw returns for comparison. The test uses the Fisher transformation $\operatorname{artanh}\hat{\rho}$ with variance $1/(T-3)$.

6.2 DCC Estimates

The maximum likelihood DCC estimates are $\hat{a} = 0.052$ (0.014) and $\hat{b} = 0.920$ (0.021) (Table 8, robust standard errors), so both parameters are precisely estimated, although the standard errors ignore the first-step estimation error and are somewhat too small. The persistence $\hat{a} + \hat{b} = 0.972$ implies a half-life of about 25 trading days for a shock to the correlation. DCC improves the log-likelihood by 42 points over a constant correlation. Under the null of constant correlation b is not identified, so the LR statistic of 84.8 has no standard distribution, but it is far too large to be explained by chance.

Table 8: DCC(1,1) estimates for the standardised GJR(1,1,1) residuals

	Estimate	Robust s.e. (s.e.)
a	0.0524	0.0136 (0.0087)
b	0.9201	0.0208 (0.0147)
$a + b$	0.9724	
$\log L$, DCC	−2856.62	
$\log L$, constant correlation ($\hat{\rho} = 0.810$)	−2899.01	
LR statistic	84.77	
Range of ρ_t	0.441 – 0.932	

Figure 5 plots ρ_t together with a 63-day rolling correlation as a model-free check. The two agree well, and DCC reacts faster. The correlation varies between 0.44 and 0.93. As a proxy for currency noise, the lower panel shows the rolling standard deviation of the return difference VUAG.L − CSPX.L, which is driven by the GBP/USD rate. Over the full sample it is negatively related to ρ_t (correlation −0.41). The first deep trough ($\rho_t \approx 0.54$ in November 2022) coincides with the sterling swings after the September 2022 mini-budget, about a month after this proxy peaked at 1.4% (still 1.2% at the trough), against a median of 0.45%. The minimum of 0.44 in December 2023 coincides with the large dollar moves after the Federal Reserve signalled rate cuts; the proxy rose only moderately then (0.60%), but equity

volatility was very low, so the currency part weighed relatively heavily. At such times the two funds behave noticeably differently, even though they hold the same stocks.

Around Liberation Day the correlation did not fall but rose, from 0.90 on 2 April to 0.93 on 4 April, just below the sample maximum (0.932 in January 2022), and it stayed at about 0.90 or higher for the rest of April. The equity shock was so large that it dominated the currency moves, and both standardised residuals were large and of the same sign on several consecutive days. The rise is modest (from 0.90 to 0.93, and we report no confidence band for ρ_t), but it goes in the direction typical of market-wide crises: diversification between the two currency lines did not increase when it would have been most useful.

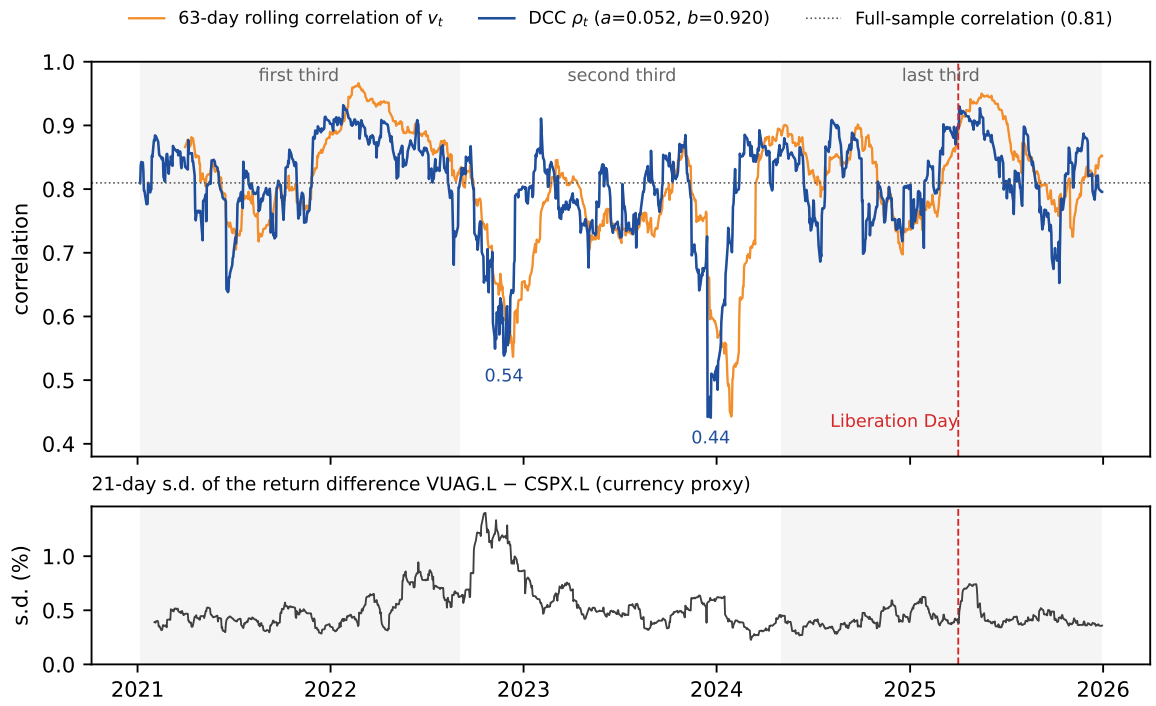


Figure 5: Dynamic conditional correlation ρ_t (CSPX.L vs. VUAG.L), with a 63-day rolling correlation and the constant full-sample correlation (top), and the 21-day standard deviation of the return difference VUAG.L − CSPX.L as a currency proxy (bottom). Shading marks the three thirds of the sample; the dashed red line is Liberation Day.

7 Conclusion

Our main findings are:

1. **GARCH(1,1)** clearly beats EWMA and RiskMetrics. The data reject $\omega = 0$ and $\alpha + \delta = 1$, and the integrated models react too little to large shocks and stay elevated for too long afterwards. RiskMetrics, despite having no estimated parameters, is almost as good as EWMA, since $\hat{\delta} \approx 0.94$.

2. **GJR(1,1,1)**, selected by BIC from 36 specifications, improves substantially on GARCH(1,1). In this model the volatility response is dominated by negative returns, and it followed the Liberation Day shock most closely. It also gives the best out-of-sample variance forecasts for 2025, significantly better than EWMA and RiskMetrics.

3. All models capture the volatility clustering, but the Gaussian models leave fat-tailed residuals: their q_t are not uniform. Student-t errors ($\hat{\nu} \approx 5$) largely fix this and give better density forecasts; only the skewness of CSPX.L remains.

4. **DCC(1,1)** shows a high but clearly time-varying correlation ($a + b = 0.97$). The correlation falls when pound/dollar volatility is high and rose during the tariff crash.

8 Limitations

1. **Distributional assumptions:** Student-t errors capture the fat tails, but not the negative skewness of CSPX.L; a skewed-t distribution would address this.

2. **Forecast design:** the parameters are estimated once and one year of forecasts, dominated by April 2025, is a short evaluation sample. Rolling re-estimation and a longer period would give more power, especially for the MSE loss.

3. **Boundary estimate of α_1:** because $\hat{\alpha}_1 = 0$, the standard errors of the other GJR parameters can be distorted. We looked into fixing $\alpha_1 = 0$, i.e. GJR(1,0,1): the estimates and the likelihood are unchanged; for VUAG.L the standard errors hardly change, but for CSPX.L they become much smaller (for the CSPX.L t model 0.04 instead of 0.29 for $\hat{\delta}_1$; ω for CSPX.L then has $t \approx 3.2$ in the normal model and 2.5 in the t model). As $q = 0$ lies outside the grid of the assignment, we keep GJR(1,1,1) and leave proper boundary inference outside the scope of this report.

4. **Two-step DCC inference:** the DCC standard errors ignore the estimation error of the GJR step and are somewhat too small.

5. **Variance proxy:** y_t^2 is a noisy proxy for the daily variance. Intraday realised volatility would sharpen both the forecast comparison and the analysis of announcement days.

AI Disclosure

This disclosure is written in accordance with the academic integrity guidelines of Vrije Universiteit Amsterdam.

Use of AI. For this assignment we used three generative AI tools: Gemini (Google), ChatGPT (OpenAI) and Claude (Anthropic, model Opus 5.5, used through the Claude Code tool). Gemini and ChatGPT were used early in the project as a coding assistant and text-formatting tool. We list the purposes below, with approximate prompts and how we checked the output.

- **Coding assistance and formatting (Gemini, ChatGPT).** These tools helped to debug the Python parameter-optimisation routines in Arnav's first implementation, to generate Matplotlib commands for exporting high-resolution figures, and to turn raw empirical output into LaTeX code.
- **Python code (Claude).** Claude did most of the coding of the notebook, under extensive monitoring by us. Approximate prompt: "Student-t errors for GJR(1,1,1) and redo the q_t histogram". We did not take the code for granted and checked that the results were sensible and consistent, for example that larger nested models never fit worse, that the estimates satisfy the parameter constraints, and that rerunning the notebook from `returns.csv` alone gives identical tables.
- **LaTeX layout and language (Claude).** Claude set up the LaTeX layout and edited our text where needed or where it was not academic enough, while keeping our tone and message the same. The substantive choices were ours, for example which ETFs to use, selecting the model by BIC, keeping α_1 free and discussing the boundary estimate instead of re-estimating, and using one degree of freedom in the LR test of RiskMetrics against EWMA.
- **Review and references (Claude).** We asked Claude to act as an independent grader and check the report against the assignment brief ("Make a sub-agent check and grade the PDF against the assignment"), and to look up and verify the references and their DOIs via Crossref. We also asked it to check whether our statistical tests were set up appropriately, for example the degrees of freedom of the LR tests; it confirmed that they were. The last checks graded the report 9.2–9.3/10. We decided ourselves which suggested changes to adopt.
- **This disclosure (Claude).** The same model drafted this section from our notes on how we used AI. We reviewed and corrected it.

Reflection. Generative AI saved us time on implementation and layout, which allowed us to spend more time on the econometric choices and their interpretation. However, code generated by AI can run without errors and still be wrong, so checking every step against the data, the assignment brief and our own understanding was essential. We remain responsible for all content of this report and can explain every result in it.

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A All GJR Models

Table 9: Fit of all 36 GJR(p, q, o) models

		CSPX.L			VUAG.L		
(p, q, o)	k	$\log L$	AIC	BIC	$\log L$	AIC	BIC
(0,1,0)	3	−1740.9	3487.9	3503.3	−1652.2	3310.4	3325.8
(0,2,0)	4	−1726.0	3460.0	3480.6	−1642.6	3293.1	3313.7
(0,3,0)	5	−1710.0	3430.1	3455.7	−1632.7	3275.4	3301.1
(1,1,0)	4	−1683.7	3375.3	3395.9	−1592.6	3193.3	3213.8
(1,2,0)	5	−1683.7	3377.3	3403.0	−1591.8	3193.6	3219.3
(1,3,0)	6	−1683.7	3379.3	3410.1	−1591.3	3194.7	3225.5
(2,1,0)	5	−1683.0	3376.0	3401.6	−1592.6	3195.3	3221.0
(2,2,0)	6	−1683.0	3378.0	3408.8	−1590.8	3193.5	3224.4
(2,3,0)	7	−1683.0	3380.0	3415.9	−1590.7	3195.4	3231.4
(3,1,0)	6	−1683.0	3378.0	3408.8	−1592.2	3196.3	3227.1
(3,2,0)	7	−1683.0	3379.9	3415.9	−1589.2	3192.5	3228.5
(3,3,0)	8	−1682.7	3381.4	3422.5	−1589.2	3194.5	3235.6
(0,1,1)	4	−1735.0	3477.9	3498.5	−1640.8	3289.6	3310.2
(0,2,1)	5	−1718.0	3446.0	3471.7	−1630.8	3271.7	3297.4
(0,3,1)	6	−1699.2	3410.4	3441.2	−1619.1	3250.2	3281.1
(1,1,1)	5	−1649.5	3309.0	**3334.7**	−1560.6	3131.2	**3156.9**
(1,2,1)	6	−1649.5	3311.0	3341.8	−1560.6	3133.2	3164.0
(1,3,1)	7	−1649.5	3313.0	3349.0	−1560.6	3135.2	3171.1
(2,1,1)	6	−1648.4	**3308.8**	3339.7	−1560.6	3133.1	3163.9
(2,2,1)	7	−1648.4	3310.8	3346.8	−1560.6	3135.1	3171.1
(2,3,1)	8	−1648.4	3312.8	3353.9	−1560.6	3137.1	3178.2
(3,1,1)	7	−1648.4	3310.8	3346.8	−1559.2	3132.5	3168.4
(3,2,1)	8	−1648.4	3312.8	3353.9	−1559.2	3134.5	3175.5
(3,3,1)	9	−1648.4	3314.8	3361.1	−1559.2	3136.5	3182.7
(0,1,2)	5	−1726.7	3463.3	3489.0	−1632.3	3274.7	3300.4
(0,2,2)	6	−1716.7	3445.4	3476.2	−1629.8	3271.5	3302.3
(0,3,2)	7	−1698.3	3410.6	3446.6	−1618.8	3251.6	3287.5
(1,1,2)	6	−1649.5	3311.0	3341.8	−1560.6	3133.2	3164.0
(1,2,2)	7	−1649.5	3313.0	3349.0	−1560.6	3135.2	3171.1
(1,3,2)	8	−1649.5	3315.0	3356.1	−1560.6	3137.2	3178.3
(2,1,2)	7	−1648.3	3310.7	3346.6	−1560.4	3134.8	3170.8
(2,2,2)	8	−1648.3	3312.7	3353.8	−1560.4	3136.8	3177.9
(2,3,2)	9	−1648.3	3314.7	3360.9	−1560.4	3138.8	3185.1
(3,1,2)	8	−1648.3	3312.7	3353.8	−1557.5	**3131.0**	3172.1
(3,2,2)	9	−1648.3	3314.7	3360.9	−1557.5	3133.0	3179.2
(3,3,2)	10	−1648.3	3316.7	3368.0	−1557.5	3135.0	3186.4

p: lags of σ_t^2, q: lags of u_t^2, o: asymmetric lags. k: number of parameters (including μ and ω). Bold: smallest AIC and BIC per series.

B Code and Data

The notebook `EQF Assignment 2 (9-10 change).ipynb` extracts the returns from the course price file `etf_full-L.csv` into `returns.csv` (Section 2), and then uses only that file to estimate all models and write every table and figure of this report to `output/`. Running all cells reproduces all results in about fifteen seconds; without the price file the notebook starts from `returns.csv` directly.